



N029AV (S/N: W14061878122) — Modem/SIM Connectivity Diagnostic Report

Immfly Ground Operations / IT — Session: August 31–September 1, 2026

Field	Value
Machine	N029AV (S/N: W14061878122)
Affected component	Cellular data connectivity (Sierra Wireless EM7565 modem / SIM)
Initial symptom	No internet connectivity following a disk re-image; modem reported sim-missing
SIM detection issue	RESOLVED — a physical SIM reseal cleared the sim-missing state
Remaining issue	UNRESOLVED — modem authenticates successfully via PAP, but IP address assignment (IPCP) never completes
Signal strength	Confirmed strong and stable (UMTS RSSI -57 to -67 dBm) throughout troubleshooting
Suspected root cause	Possible carrier-side data provisioning issue on this specific SIM, rather than a device/hardware fault
Recommended action	Escalate to Movistar (via Luisa/carrier contact) to verify data-plan provisioning and APN activation for this SIM's ICCID

Narrative Summary

N029AV lost cellular connectivity following a disk re-image. Initial diagnosis via `mmcli` showed the modem in a **failed** state with **failed reason: sim-missing**, despite the modem's firmware and USB mode (QMI) both being up to date. Two separate issues were identified and investigated over the course of this troubleshooting session, spanning two days.

Issue 1 (resolved): SIM detection. A udev rule correction was first applied to ensure the AT command port was correctly exposed to ModemManager (the dynamic `blinterfaceNumber`-to-port mapping for this fleet varies by unit and by boot, and is expected behavior — ModemManager's own reported port list was cross-checked against the `modemAT/modemPPP` symlinks each time to confirm correctness). A logical modem reset (`mmcli -m 0 --reset`) intermittently cleared the `sim-missing` state on one occasion, but the condition recurred after a subsequent reboot. A full physical SIM reseal (power off, remove and reinsert the SIM, power on) was performed and definitively resolved the `sim-missing` state going forward — the modem consistently reported the SIM as present and transitioned to a normal *disabled/enabled* state afterward.

Issue 2 (unresolved): network registration / IP assignment. With the SIM detection issue resolved, the modem could be enabled successfully and consistently showed strong, stable signal (UMTS RSSI between -57 dBm and -67 dBm across multiple checks, confirmed via `mmcli --signal-get`) — ruling out an antenna or RF coverage problem. However, ModemManager's own 3GPP registration state remained stuck at **idle** indefinitely, never progressing to *registered*, even after forcing the allowed mode to 3G-only, cycling disable/enable repeatedly, and explicitly forcing registration with the known Movistar Colombia operator ID (732123) via `mmcli`

--3gpp-register-in-operator — all attempts at forced registration through ModemManager/QMI resulted in a timeout.

A separate, lower-level test was then performed by launching the classic PPP dial-up connection directly (via pon/chat, the mechanism this older-firmware unit uses independently of ModemManager's own QMI registration state). This test produced a more informative result: **LCP negotiation completed successfully, and PAP authentication succeeded** — meaning the SIM's credentials were accepted by the carrier network at the PPP session level. However, the subsequent **IPCP negotiation never completed**: the client repeatedly sent IPCP ConfReq messages requesting an IP address and DNS servers, but received no response from the network side, eventually timing out ("IPCP: timeout sending Config-Requests") and terminating the connection.

This is a meaningful distinction: successful PAP authentication indicates the SIM is recognized and accepted by the carrier for network access, while a failure specifically at the IP-assignment (IPCP) stage — with strong, stable signal confirmed throughout — points toward the data session / APN not being correctly provisioned for this SIM on the carrier side, rather than a device-level or hardware fault in N029AV itself.

Troubleshooting Steps Performed

1. Initial diagnosis

modem_fix.py / mmcli confirmed firmware and USB mode (QMI) up to date, but modem state: failed, failed reason: sim-missing.

2. udev symlink verification

Confirmed modemAT/modemPPP symlinks matched the port ModemManager itself reported as the AT port (cross-checked via 'mmcli -m 0' System/ports section) — port-to-symlink mapping was correct.

3. Logical modem reset

mmcli -m 0 --reset cleared the sim-missing state on the first day of troubleshooting; modem reached 'disabled' state and was successfully enabled, but registration remained stuck at 'idle' despite strong confirmed signal.

4. Forced registration attempts (day 1)

Attempted mmcli -m 0 --set-allowed-modes="3g" (isolating LTE as a variable) and mmcli -m 0 --3gpp-register-in-operator=732123 (explicit Movistar Colombia operator ID). Both the 3G-only mode and the explicit operator registration attempt resulted in a 'Timeout was reached' error.

5. Session paused

Given the time invested and the need to work other units, troubleshooting was paused for the day with the issue unresolved.

6. Reboot and sim-missing recurrence (day 2)

After a subsequent reboot, mmcli again reported failed reason: sim-missing, despite the previous day's logical reset having cleared it — confirming the logical reset was not a durable fix.

7. Physical SIM reseal

Powered off the unit, physically removed and reinserted the SIM, and powered back on. The modem consistently reported 'disabled' state afterward (no sim-missing), and successfully transitioned to 'enabled' when commanded — this is treated as the confirmed fix for the SIM-detection issue.

8. Signal verification

mmcli --signal-setup=10 followed by --signal-get confirmed strong UMTS signal (RSSI -57 dBm, Ec/Io -4.50 dB), ruling out an RF/antenna cause for the still-unresolved registration issue.

9. Registration state re-check

Despite the resolved SIM detection and confirmed strong signal, mmcli continued to report state: enabled / registration: idle — the modem never progressed to a registered state via ModemManager/QMI.

10. Direct PPP dial-up test

Launched the classic chat/pppd dial-up connection directly (independent of ModemManager's QMI registration state, as this older-firmware unit relies on this mechanism). LCP negotiation and PAP authentication both succeeded, but IPCP (IP address/DNS assignment) never completed, timing out after repeated unanswered Config-Request messages.

Conclusion and Recommendation

N029AV's SIM-detection issue has been resolved via a physical SIM reseal. The remaining connectivity issue — successful carrier authentication (PAP) but failure at the IP-assignment (IPCP) stage, with strong and stable signal confirmed throughout — is not consistent with a device-level hardware or firmware fault. It is consistent

with a carrier-side data-session provisioning problem specific to this SIM (for example, the associated data plan/APN not being correctly activated or provisioned for this ICCID on Movistar's network).

Recommended action: Escalate this specific SIM/ICCID to Movistar (via the established carrier contact) to verify that data-plan provisioning and APN activation are correctly configured for this SIM. No further device-level troubleshooting is recommended for N029AV's modem itself unless the carrier confirms provisioning is correct and the issue persists, in which case a SIM swap (testing with a known-good SIM from another unit) would be the next logical diagnostic step to conclusively separate a SIM-provisioning cause from a modem-hardware cause.

Appendix: Key PPP Session Log

```
PAP authentication succeeded
sent [CCP ConfReq id=0x1 ]
sent [IPCP ConfReq id=0x1 ]
rcvd [LCP ProtRej id=0x1b ...] (Compression Control Protocol rejected -- benign)
sent [IPCP ConfReq id=0x1 ...] (repeated, no response received)
sent [IPCP ConfReq id=0x1 ...] (repeated, no response received)
sent [IPCP ConfReq id=0x1 ...] (repeated, no response received)
IPCP: timeout sending Config-Requests
sent [LCP TermReq id=0x2 "No network protocols running"]
Connection terminated.
Modem hangup

Signal at time of test:
  UMTS | rssi: -57.00 dBm
       | ecio:  -4.50 dB
```